

# Literature Review

## Overview

This review surveys recent advances in zero-shot speech synthesis, focusing on how modern systems generate unseen speakers or styles without speaker-specific fine-tuning. It examines three complementary strands of research: (1) zero-shot adaptation pipelines that combine robust speaker encoders with flexible conditioning mechanisms, enabling any-to-any voice conversion—including whispered or expressive speech; (2) the development and refinement of speaker-embedding representations, where newer architectures such as ECAPA-TDNN are benchmarked against traditional x-vectors and further enhanced by residual adapters that narrow the synthetic-natural embedding gap; and (3) architectural innovations that permit dynamic, inference-time adaptation, ranging from dual-style encoding networks to lightweight residual MLP adapters stacked on frozen encoders. By integrating these themes, the review aims to synthesize a unified picture of how embedding quality, adaptation strategies, and modular network designs collectively push zero-shot TTS toward more natural, speaker-agnostic synthesis.

## Thematic Grouping

*Zero-shot speaker/style adaptation in speech synthesis* - The three studies converge on the challenge of generating speech from speakers or styles never encountered during training. (YourTTS) builds a zero-shot pipeline that couples a pre-trained ECAPA-TDNN speaker encoder with a diffusion-based acoustic model, while (DS-TTS) extends this idea by introducing a dual-style encoder that simultaneously captures timbre and prosodic cues. (SpeakerVC) further generalises the paradigm to any-to-any voice conversion, demonstrating that the same encoder-decoder backbone can be repurposed for whispered speech without additional fine-tuning. Collectively, these works illustrate a shift from speaker-specific adaptation toward universal, encoder-driven synthesis.

*Speaker embedding representations and enhancement* - This theme centres on improving the fidelity of speaker embeddings and narrowing the synthetic-natural gap. (An Exploration of ECAPA-TDNN) provides a systematic comparison of x-vector and ECAPA-TDNN encoders, establishing ECAPA-TDNN as a superior source of speaker identity. (Improvement Speaker Similarity for) leverages that encoder within a conversion framework, showing that higher-quality embeddings translate directly into more speaker-consistent outputs. (VOCALIS\_preprint) complements these findings by introducing a residual adapter that refines the raw ECAPA-TDNN vectors, effectively bridging the distributional mismatch between synthetic and natural speech. Together, the papers map a clear trajectory: from selecting robust encoders to actively enhancing their representations for downstream synthesis.

*Architectural innovations for dynamic or residual adaptation* - Both (DS-TTS) and (VOCALIS\_preprint) propose lightweight modules that modify speaker representations at inference time rather than retraining the entire model. (DS-TTS) embeds a Dual-Style Encoding Network and a Dynamic Generator Network that condition generation on mutable style codes, enabling on-the-fly adaptation to new timbres and prosodies. In parallel, (VOCALIS\_preprint) adds a residual MLP adapter atop a frozen speaker encoder, offering a plug-and-play mechanism to correct embedding drift. The synergy between

these approaches underscores a growing emphasis on modular, runtime-adjustable architectures that preserve the core acoustic model while flexibly accommodating new speaker characteristics.

## Methodology Comparison

| Paper                                         | Method / Approach                                                                                                                                                    | Key Technique                                                                                                                                                                  | Evaluation Approach                                                                                                                                            |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>An Exploration of ECAPA-TDNN.pdf</b>       | Integrates three speaker encoders (H/ASP baseline, x-vector, ECAPA-TDNN) into the YourTTS zero-shot multi-speaker TTS framework.                                     | Swaps the source of speaker embeddings in a VITS-based YourTTS model to assess representation quality.                                                                         | Subjective listening test for speaker similarity + objective cosine-distance between reference and synthesized embeddings.                                     |
| <b>DS-TTS.pdf</b>                             | Proposes DS-TTS, a zero-shot voice-cloning system that extracts speaker style via a Dual-Style Encoding Network and renders speech with a Dynamic Generator Network. | Dual-Style Encoding Network (DuSEN) separates timbre and prosody; Dynamic Generator Network (DyGN) modulates phoneme representations with dynamic variance adaptation.         | MOS for naturalness, speaker-similarity scores, and objective metrics on LibriTTS & VCTK, compared against recent zero-shot baselines.                         |
| <b>Improvement Speaker Similarity for.pdf</b> | Introduces SpeakerVC, a zero-shot any-to-any voice-conversion system built on StyleTTS2 that works for regular and whispered speech.                                 | HuBERT encoder (streaming-adapted) + ECAPA-TDNN speaker loss; two decoder variants (Tacotron-2 and FastSpeech-2) with ECAPA-TDNN-driven speaker conditioning.                  | Subjective SMOS ratings via Toloka; objective WER, EER, and SIM-o to measure speaker similarity and intelligibility across domains.                            |
| <b>VOCALIS_preprint.pdf</b>                   | Presents VOCALIS, a lightweight residual adapter that maps synthetic speaker embeddings to the natural-speech embedding space.                                       | Frozen pretrained ECAPA-TDNN encoder + two-layer MLP residual adapter trained with a composite loss (cosine similarity, MSE, regularization) on paired synthetic-natural data. | Cosine similarity between adapted synthetic embeddings and natural references; downstream TTS speaker-similarity improvement without retraining the TTS model. |

### Notes

- The methodology of *Improvement Speaker Similarity for.pdf* blends voice conversion (VC) and TTS techniques, which makes it slightly broader than a pure TTS cloning approach; the table captures its core components (encoder, loss, decoder) while the prose clarifies this hybrid nature.

## Key Findings Comparison

Across the four studies, a common thread is the quest to close the gap between synthetic and natural speaker representations, yet the papers diverge in how they achieve--or fail to achieve--this goal.

Both **(An Exploration of ECAPA-TDNN.pdf)** and **(VOCALIS\_preprint.pdf)** agree that the choice of speaker encoder critically shapes zero-shot cloning quality. The former shows that the older H/ASP encoder surprisingly outperforms newer x-vector and ECAPA-TDNN embeddings when plugged directly into the YourTTS framework, suggesting that raw encoder performance does not automatically translate to better TTS. Complementing this, VOCALIS demonstrates that a lightweight residual adapter can *post-process* ECAPA-TDNN embeddings to bring them closer to the natural-speech space, thereby rescuing the otherwise sub-optimal performance observed in the former study.

In contrast, **(DS-TTS.pdf)** and **(Improvement Speaker Similarity for.pdf)** disagree with the implication that a single encoder suffices. DS-TTS introduces a dual-style encoding (timbre++prosody) and a dynamic generator, reporting higher speaker similarity--especially for short utterances--than prior

zero-shot baselines. SpeakerVC goes further by integrating ECAPA-TDNN-driven speaker loss within a StyleTTS2 backbone and by handling whispered speech, achieving superior similarity and intelligibility across domains. These results complement DS-TTS by confirming that richer, style-aware representations and targeted loss functions can overcome the limitations highlighted by the encoder-only experiments in **(An Exploration of ECAPA-TDNN.pdf)**.

Overall, the literature converges on the importance of *adaptation*--whether through residual adapters (VOCALIS) or dual-style encodings (DS-TTS, SpeakerVC)--while disagreeing on whether a vanilla, high-capacity encoder alone can deliver state-of-the-art zero-shot cloning. The complementary findings suggest a promising hybrid direction: combine robust encoders with lightweight adaptation modules to maximize speaker similarity without retraining the entire TTS system.

## Gaps & Future Directions

Despite impressive strides, the current body of work leaves several critical questions unanswered. First, none of the studies evaluate **cross-lingual zero-shot cloning**, leaving open whether the encoder-plus-adapter pipelines (e.g., VOCALIS-(-VOCALIS\_preprint.pdf) or DS-TTS-(-DS-TTS.pdf)) generalise to languages unseen during encoder training. Second, the **robustness to noisy or low-resource reference audio** remains unexplored; both (-An Exploration of ECAPA-TDNN.pdf) and (-Improvement Speaker Similarity for.pdf) assume clean recordings, yet real-world applications often rely on telephone or far-field captures. Third, while dual-style encodings improve short-utterance similarity, **long-form coherence and prosodic consistency** have not been systematically measured across the zero-shot systems. Fourth, the **computational trade-offs of dynamic adapters** versus static encoders are only qualitatively discussed, lacking rigorous latency and memory benchmarks for on-device deployment. Finally, the **ethical implications of near-perfect voice cloning**--including speaker consent, misuse detection, and watermarking--are absent from all four papers. Addressing these gaps will require multilingual, noisy-robust datasets, long-form evaluation protocols, detailed efficiency analyses, and integrated safeguards to guide responsible zero-shot TTS deployment.

## References

- An Exploration of ECAPA-TDNN.pdf
- DS-TTS.pdf
- Improvement Speaker Similarity for.pdf
- VOCALIS\_preprint.pdf